

Summary: Completed Work

Inside Airbnb - New York City | Supplementary Deliverable (Assignment Section 11.1)

Strategy	Single city (New York City), depth-focused over breadth
Fully Completed	Section 02 (Mandatory) · Section 03.1-3.5 (Recommended) · Section 04 (Recommended) · Final Report
Partially / Not Completed	Section 03.6 (Optional) · Sections 05–08 -see companion <i>Summary: Incomplete Work</i>

This document summarises what was implemented in this submission and to what standard, so a reviewer can verify scope and quality without reading the full report or notebooks line by line. The guiding strategy was deliberate: invest deeply in the mandatory and highest-rubric-weight sections (Dataset Familiarization and Data Engineering), and use the remaining time to convert that engineering work into business-facing analysis (EDA), rather than spreading effort thinly across every optional section.

At a Glance

Section	Category	Status	Standard Achieved
02 - Dataset Familiarization	Mandatory	Complete	Full schema documentation, PK/FK mapping, data-dictionary cross-reference, limitations and business-domain framing across all 4 files used.
03.1 - Ingestion & Profiling	Recommended	Complete	Repeatable, idempotent pipeline; full profiling, duplicate detection, completeness assessment, outlier identification, domain validation.
03.2 - Cleaning & Standardization	Recommended	Complete	Price, date, boolean and categorical standardisation; 3 missing-value strategies evaluated and one selected with documented rationale.
03.3 - Enrichment & Joining	Recommended	Complete	Enriched listing-master table built from listings + reviews + calendar + neighbourhood aggregates, with 7 derived fields.
03.4 - Data Modeling	Recommended	Complete	Star schema designed and implemented in DuckDB (2 fact tables, 4 dimension tables) with 6 analytical SQL queries demonstrated.
03.5 - Pipeline Automation	Recommended	Complete	Structured logging, exponential back-off retries, MD5-based incremental processing, SQLite metadata layer, and a full data-lineage table.
03.6 - Advanced/Cloud-Native	Optional	Not attempted	See Summary: Incomplete Work.
04.1-4.5 - EDA (all 5 subsections)	Recommended	Complete	Distributions, geographic/spatial analysis, temporal trends, host/supply-side and review/demand-side analysis — every finding paired with a business interpretation.
05-08 -Statistics / ML / AI / Innovation	Recommended & Optional	Partially designed	See Summary: Incomplete Work.
Final Report (25 pp.)	Mandatory	Complete	All 15 report sections plus Appendix A, professionally formatted with numbered sections, table of contents and consistent typography.

Section 02 - Dataset Familiarization

Completed in full, in *01_dataset_familiarization.ipynb*. The schema of all four files used (listings.csv.gz, calendar.csv.gz, reviews.csv.gz, neighbourhoods.csv) was documented column by column, the entity-relationship structure was mapped (*listings.id* as the primary key referenced by calendar and reviews), 7 dataset limitations were identified (12 fully-null scraping failure columns, no historical pricing, 8,744 missing prices, point-in-time snapshot, orphaned records, no guest data), and the business meaning of each entity (listing, host, review) was framed in plain language before any cleaning began.

Section 03 - Data Engineering Challenges (3.1–3.5)

- **Ingestion & Profiling (3.1):** a skip-if-exists, retrying download pipeline loaded all 4 files (30,259 listings × 90 columns; 11.2M calendar rows; 990,170 reviews; 230 neighbourhoods). A custom quality-scoring framework graded each file across completeness, validity, uniqueness and consistency - listings.csv scored 84.9/100 (Grade B); calendar, reviews and neighbourhoods all scored 100/100 with zero duplicates.
- **Cleaning & Standardization (3.2):** price strings (“\$113.97”) converted to nullable floats (21,515 valid records); 5 date columns parsed to UTC-aware datetimes with 0 future-date violations; boolean flags normalised from 't'/'f' strings; 69 free-text property types collapsed into 5 canonical groups; 3 missing-value strategies tested for bedrooms/bathrooms/beds, with group-median imputation selected and flagged transparently.
- **Enrichment & Joining (3.3):** listings joined with review summaries and calendar-derived occupancy/revenue estimates; neighbourhood-level aggregates computed; 7 derived fields built (occupancy_rate, estimated_revenue, has_reviews, review_frequency, price_per_bedroom, host tenure, neighbourhood_median_price); a cross-city scaffold was added for future multi-city extension.
- **Data Modeling (3.4):** a star schema was designed and implemented in DuckDB - *fact_listing* (21,515 rows) and *fact_calendar* (11,152,576 rows) against *dim_host*, *dim_neighbourhood*, *dim_property_type* and *dim_date*. Six analytical SQL queries were written and run against it (revenue by room type, top neighbourhoods by price, weekday/weekend booking pressure, monthly seasonality, top host portfolios, tenure vs. price).
- **Pipeline Automation (3.5):** structured console + file logging, exponential back-off retry logic (2s/4s/8s, capped at 60s), MD5-checksum-based incremental processing, a SQLite metadata layer recording every pipeline run, and a documented data-lineage table tracing every output table back to its source files and transformations.

Section 04 - Exploratory Data Analysis (all subsections)

Completed in full, in *04_Exploratory_Data_Analysis.ipynb*, with every finding accompanied by the business interpretation the assignment brief requires - not just a technical description. Highlights:

- **4.1 Summary Statistics & Distributions:** descriptive statistics for all key numerical variables; price distribution (median \$174.69 vs. mean ~\$240, right-skewed); host listing-count power-law dynamics; review-score rating inflation; availability pattern bimodality.
- **4.2 Geographic & Spatial Analysis:** listing-density mapping across 222 neighbourhoods (top 3 hold 18.3% of supply); a geographic pricing gradient (Pearson $r = -0.307$ vs. distance from Times Square); spatial review-score patterns; property-type geographic clustering.
- **4.3 Temporal & Seasonal Trends:** 17-year review-volume growth (41 reviews in 2009 to 148,331 in 2023, with a clear 2020 COVID dip and rapid recovery); the discovery that median minimum stay is 30 nights city-wide - correctly traced to NYC's Local Law 18, not a data artifact.
- **4.4 Host & Supply-Side Analysis:** three-tier host segmentation (casual / small professional / large commercial) with portfolio-size vs. price relationships, including a Kruskal-Wallis test across host segments and Mann-Whitney U tests comparing Superhost vs. non-Superhost listings on rating and price.
- **4.5 Review & Demand-Side Analysis:** the review-count/price/score relationship; identification of a “high demand, low quality” listing segment (many reviews, sub-4.0 scores); review sub-dimension (cleanliness, communication, location) comparison.